

HemoSight

Federated Edge-AI for Peripheral Blood Smear Screening in Rural Primary Health Centres

M Sujith Sali · ISE Dept, VTU Karnataka · Apache-2.0

Portfolio & Innovation Guide — 2027 track

SCREENING AID ONLY — NOT A DIAGNOSTIC DEVICE.
Flags abnormalities with calibrated confidence; never diagnoses disease.

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HemoSight produces abnormality flags with calibrated confidence and a needs-review flag. It does not diagnose disease and never emits a disease name. Every result must be confirmed by a qualified pathologist or physician. All benchmark numbers are on public datasets (Raabin-WBC, NIH Malaria) that are off-domain from Indian PHC settings and require [TARGET]-tagged clinical validation before any real-world use.

HemoSight

Open-source federated-learning framework for on-device blood microscopy screening in Indian PHC settings, evaluated on Raabin-WBC and NIH Malaria public benchmarks.

Author: **M Sujith Sali**, ISE Dept, VTU Karnataka · License: Apache-2.0

HemoSight runs on a Raspberry Pi 5 with a USB microscope. It (1) detects blood cells with YOLOv8, (2) classifies WBCs into 5 clinical classes, (3) flags malaria parasites, (4) quantifies uncertainty with MC-Dropout, (5) federates model updates across simulated PHCs with Flower (FedAvg/FedProx) **without sharing patient images**, (6) offers a Kannada/Hindi/English UI with voice prompts, and (7) generates cryptographically signed PDF reports.

Integrity guarantees (enforced in code)

- **No fabricated metrics.** Numbers come only from logged MLflow runs with a fixed seed and dataset hash. Un-run metrics read PENDING with the exact make target — never a placeholder value.
- **Provenance tagging.** Every metric is [BOOTSTRAP] (public, off-domain) or [TARGET] (real PHC samples, not yet collected). Never blended silently.
- **No cloud in the inference path.** `common/egress_guard.no_egress()` monkey-patches `socket.socket` and raises on any outbound connection during `predict()`. Federated aggregation is the only sanctioned network path and it transmits weight deltas only.
- **Reproducibility.** `common/seed_everything.py` seeds Python/NumPy/PyTorch (CPU+CUDA), sets `cupnn.deterministic`, and pins `PYTHONHASHSEED`.

Benchmark numbers

All values below are **PENDING** until the corresponding training run is executed; the code logs real results to MLflow at that point.

Task	Dataset	Metric	Value	Provenance	Run target
Cell detection	BCCD	mAP@50 / mAP@50-95	PENDING	[BOOTSTRAP]	make train-yolo
WBC 5-class	Raabin-WBC	macro-F1 (95% CI)	PENDING	[BOOTSTRAP]	make train-wbc
WBC calibration	Raabin-WBC	ECE / Brier (pre/post)	PENDING	[BOOTSTRAP]	make train-wbc
Malaria binary	NIH Malaria	ROC-AUC / PR-AUC	PENDING	[BOOTSTRAP]	make train-malaria
Federated	Raabin (non-IID)	global acc/round	PENDING	[BOOTSTRAP]	make federated-simulate
Edge latency	Pi 5 8GB	s/image (target <3s)	PENDING	[TARGET]	run edge/infer.py on Pi

Quickstart

```
make bootstrap          # install deps + pre-commit
make download-data      # BCCD/NIH via script; Raabin needs manual registration
make test               # 20 tests, 80% coverage gate on ml/ and api/
make train-wbc          # populates the WBC row above (needs the dataset)
make federated-simulate # 5 non-IID PHC clients, FedAvg + FedProx
docker compose up --build
```

Hardware bill of materials (indicative INR, 2026 — verify before buying)

- Raspberry Pi 5 8GB — genuine boards from Robocraze / Silverline / ThinkRobotics / The Engineer Store (~₹7,500–9,000).
- Raspberry Pi 27W USB-C PD power supply (~₹1,500).
- Raspberry Pi Active Cooler (~₹600).
- 64/128GB Class-10 U3 microSD — Samsung EVO Plus / SanDisk Ultra (~₹700–1,300).
- Official Pi 5 case (~₹900).
- USB digital microscope 40x–1000x, 8 LEDs, metal stand, **UVC-compatible** — e.g. Microware / iBell IBL-ML1000X / Ruhza / Techie&Trendy on Amazon India (~₹1,200–3,500). **Verify UVC before purchase.**
- Prepared blood-smear teaching slides from a medical supply house (demo only — no patient data, no ethics approval needed for commercial training slides).
- *Optional:* Raspberry Pi AI Kit (Hailo-8L, 13 TOPS); 7-inch touchscreen for kiosk mode.

Citations

- Kouzehkanan et al. (2022), *Scientific Reports* — Raabin-WBC. <https://raabindata.com> · Mendeley snkd93bnjr.
- Rajaraman et al. (2018), *PeerJ* — NIH Malaria Cell Images. <https://lhncbc.nlm.nih.gov>
- BCCD dataset — https://github.com/Shenggan/BCCD_Dataset
- Ultralytics YOLOv8 — <https://github.com/ultralytics/ultralytics>
- Flower (flwr) — <https://flower.ai>
- AI4Bharat IndicTrans2 / IndicTTS — <https://ai4bharat.org>

See MODEL_CARD.md for intended use, limitations, and out-of-scope uses, and INTERVIEW_HANDBOOK.md for the placement-defence walkthrough.

HemoSight — Installation Guide (Hardware + Software)

Author: **M Sujith Sali**, ISE Dept, VTU Karnataka.

SCREENING AID ONLY — NOT A DIAGNOSTIC DEVICE. Setup instructions below do not make HemoSight a medical device. Real clinical use needs [TARGET] validation, ethics approval, and clinician oversight.

Part A — Software installation

A1. Prerequisites

- Python 3.11+ and git.
- Node.js 18+ (only for the web UI).
- On Linux, native libs for OpenCV + WeasyPrint PDF: `bash sudo apt-get update && sudo apt-get install -y \ libgll-mesa-glx libglib2.0-0 libpango-1.0-0 libpangocairo-1.0-0 \ libgdk-pixbuf-2.0-0 libffi-dev`

A2. Get the code and install

```
git clone <your-repo-url> hemosight # or unzip HemoSight.zip
cd hemosight
python -m venv .venv && source .venv/bin/activate
pip install -e ".[dev]"
```

A3. Make it fully functional (real weights, no manual data)

```
python -m scripts.train_toy # builds synthetic data + trains TinyCNNs (CPU, ~1
min)
# -> results/wbc_tiny.pt, malaria_tiny.pt (+ .onnx)
```

This produces **real model weights** so the whole pipeline runs. (Synthetic data => accuracy is not clinical; see MODEL_CARD.)

A4. Run the tests (proves the software works)

```
pytest -q # 22 tests pass (determinism, schema, egress, quality gate,
# MC-Dropout, attention gate, ONNX parity, signed PDF,
# audit chain, federated smoke, full end-to-end pipeline)
```

A5. Run the API and analyze an image

```
uvicorn api.main:app --host 127.0.0.1 --port 8000
# In another shell – get a token in Python:
# from api.auth import create_token; print(create_token("asha01"))
curl -H "Authorization: Bearer <TOKEN>" -F "file=@smear.png" \
    http://127.0.0.1:8000/analyze
# -> JSON with cell counts, WBC differential, malaria flag, uncertainty,
#     attention-gate status, and a path to a signed PDF report.
```

A6. Run the web UI (Kannada / Hindi / English)

```
cd web && npm install && npm run dev    # http://localhost:5173
```

A7. Optional — real datasets for real accuracy

make download-data (BCCD + NIH auto; Raabin needs a one-time site registration), then make train-wbc / train-malaria / train-yolo / federated-simulate / eval-all. These log real metrics to MLflow and replace the synthetic weights.

A8. Docker (everything at once)

```
docker compose up --build    # api + mlflow + postgres + aggregator
```

Part B — Hardware assembly (Raspberry Pi 5)

B1. Bill of materials (see README for Indian sellers / indicative INR)

Raspberry Pi 5 8GB · 27W USB-C PD supply · Active Cooler · 64/128GB U3 microSD · official case · UVC USB microscope (40x–1000x) · prepared teaching smear slides. Optional: Hailo-8L AI Kit, 7-inch touchscreen.

B2. Flash the OS

1. Download **Raspberry Pi Imager** on your laptop.
2. Choose **Raspberry Pi OS (64-bit, Bookworm)** → your microSD.
3. In Imager settings: set hostname hemosight, enable SSH, set username/password, configure Wi-Fi. Write.

B3. Assemble

1. Fit the **Active Cooler** onto the Pi 5 (it clips + plugs into the fan header).
2. Seat the Pi in the case, insert the flashed microSD.
3. Plug the **UVC USB microscope** into a USB 3.0 (blue) port.
4. Connect the **27W USB-C** supply last (this powers on).

B4. First boot + software

```
ssh hemosight@hemosight.local
sudo apt-get update && sudo apt-get install -y python3-venv git \
    libgl1-mesa-glx libgl1-mesa-dri libpango-1.0-0 libpangocairo-1.0-0
git clone <your-repo-url> /opt/hemosight && cd /opt/hemosight
python3 -m venv .venv && source .venv/bin/activate
pip install -e "." # ONNX Runtime path is edge-friendly
python -m scripts.train_toy # or copy trained .onnx from your dev machine
```

B5. Verify the microscope

```
python -c "import cv2; c=cv2.VideoCapture(0); ok,_=c.read(); print('camera ok:', ok)"
```

Set `HEMOSIGHT_CAMERA=uvic` (default) or `picamera` for the Pi Camera.

B6. Run as a service (auto-start on boot)

```
sudo cp edge/hemosight.service /etc/systemd/system/
sudo systemctl daemon-reload
sudo systemctl enable --now hemosight
systemctl status hemosight # should be active (running)
```

The service serves the API on `127.0.0.1:8000`. Open the web UI (or a browser on the Pi in kiosk mode with the 7-inch touchscreen) to capture and screen a smear. The inference path runs **fully offline** — the egress guard blocks any outbound call except the sanctioned federated weight-delta sync.

B7. Benchmark latency on the Pi

```
python -c "from edge.infer import EdgeInferencer, benchmark; import cv2; \
img=cv2.imread('smear.png'); print('median s/img:', benchmark(EdgeInferencer(), \
img))"
```

Report the **actual** number (target < 3 s/image); do not assume it.

Troubleshooting

- **WeasyPrint import error** → install the `libpango/libgdk-pixbuf` packages in A1/B4.
- **camera ok: False** → try index 1/2, confirm the scope is UVC, check `lsusb`.
- **ONNX provider warning** → XNNPACK falls back to CPU automatically; still works.
- **401 from /analyze** → include a valid Authorization: Bearer <token> header.

Model Card — HemoSight

SCREENING AID ONLY — NOT A DIAGNOSTIC DEVICE. Output is always an abnormality flag + calibrated confidence + needs-review flag, never a disease name. Confirm every result with a qualified clinician.

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Intended use

Assist ASHA / PHC workers in **triaging** peripheral blood smears in resource-limited rural Indian primary health centres, flagging smears that warrant review by a qualified pathologist. It is a pre-screening aid to prioritise referrals — not a substitute for laboratory diagnosis.

Models

Component	Architecture	Dataset	Provenance
Cell detection	YOLOv8n/m	BCCD	[BOOTSTRAP]
WBC 5-class	EfficientNet-B0 + MC-Dropout	Raabin-WBC	[BOOTSTRAP]
Malaria binary	MobileNetV3-Small + MC-Dropout	NIH Malaria	[BOOTSTRAP]

Uncertainty via MC-Dropout (10 passes). Calibration via temperature scaling, reported as ECE + Brier pre/post. An **attention gate** cross-checks Grad-CAM against detected cell masks ($\text{IoU} < 0.3 \Rightarrow \text{ATTENTION_MISALIGNMENT} \Rightarrow \text{needs review}$).

Metrics

All metrics are **PENDING** until the training runs complete; each is logged to MLflow with seed, git SHA, and dataset hash, and tagged [BOOTSTRAP]. No number appears here until it exists in an MLflow run.

Training-data limitations

Raabin-WBC and NIH Malaria are collected in **specific geographies with specific stains and microscopes**. Colour, staining, and optics in an Indian PHC will differ. Reported [BOOTSTRAP] performance is therefore an **upper bound in-distribution** and does **not** transfer to PHC conditions without [TARGET]-tagged validation on locally captured, ethically consented samples.

Known failure modes

- Out-of-distribution staining / illumination (mitigated partly by flat-field + gray-world preprocessing and the quality gate, but not eliminated).
- Overlapping / clumped cells degrade detection and counts.
- Low parasitemia may fall below the flagging threshold.
- MC-Dropout is an approximation of epistemic uncertainty, not a guarantee.
- The attention gate reduces, but does not remove, silent misclassification.

Out-of-scope uses

- Any autonomous diagnosis or treatment decision.
- Any use on real patients without local [TARGET] validation, ethics approval, and clinician sign-off.
- Detecting conditions outside the trained WBC classes / malaria flag.

Ethical & regulatory notes

No PII is stored by default (sample ID only). Local artifacts are encrypted (AES-256-GCM); the audit ledger is append-only and hash-chained. Design intent aligns with India's DPDP Act 2023 data-minimisation principles. HemoSight is a research/education portfolio project and is **not** a certified medical device.

HemoSight — Interview Defence Handbook

Author: **M Sujith Sali**, ISE Dept, VTU Karnataka. For 2027 placements (TCS, Infosys, Wipro, Cognizant, Accenture, JSW, product companies) and SIH / IEEE / KSCST tracks.

One-line pitch: *"HemoSight is an offline-first federated edge-AI screening aid for blood smears in rural PHCs — it flags abnormalities with calibrated confidence, never diagnoses, keeps patient images on-device, and shares only model weight deltas across clinics."*

1. System-design walkthrough (whiteboard order)

1. **Capture** — Pi 5 + UVC microscope; camera adapter is hot-swappable (Picamera2 or OpenCV VideoCapture) via config.
2. **Quality gate** — Laplacian-variance blur check, brightness window, resolution floor. Returns a *coded* error, never a bare exception.
3. **Preprocess** — fixed order: flat-field → NLM denoise → CLAHE(2.0) → gray-world WB. Deterministic for a given input.
4. **Detect** — YOLOv8 (ONNX + XNNPACK on Pi) → cell bboxes.
5. **Classify** — EfficientNet-B0 (WBC) / MobileNetV3-Small (malaria), with MC-Dropout for uncertainty.
6. **Calibrate** — temperature scaling; report ECE + Brier.
7. **Attention gate** — Grad-CAM vs cell-mask IoU; misalignment ⇒ needs-review.
8. **Serve** — FastAPI, SSE stream of the 7 pipeline stages; React UI.
9. **Report** — WeasyPrint PDF, Ed25519-signed, hash-chained audit ledger.
10. **Federate** — Flower FedAvg/FedProx over weight deltas; images never leave the device.

2. STAR answers (compressed)

Situation Rural PHCs lack pathologists; smears wait days or travel far. **Task** Build an offline screening aid that respects patient privacy and never overclaims. **Action** Designed the 10-stage pipeline above; enforced integrity in code (egress guard, provenance tags, PENDING-not-fake metrics, determinism); made it federated so clinics improve a shared model without sharing images. **Result** A reproducible, test-covered (20 tests, 80% gate), Dockerised system with signed reports and a clear screening-aid boundary — ready for [TARGET] validation.

3. Decision rationale (the questions they will ask)

Why FedAvg/FedProx, not centralised? PHC images are sensitive and bandwidth is scarce. Federation keeps images local; only weight deltas move. **FedProx** adds a proximal term so heavily **non-IID** per-PHC caseloads don't pull the global model apart — which centralised training never has to handle because it sees the pooled distribution.

Why MC-Dropout, not deep ensembles? On a Pi 5 with ~4GB usable RAM, storing and running N independent models is too costly. MC-Dropout gives an epistemic uncertainty estimate from a **single** model via N stochastic passes — cheap, and good enough to drive a needs-review flag.

Why temperature scaling? Neural nets are typically over-confident. A single scalar T fit on validation NLL fixes calibration **without changing accuracy** (it's monotonic in the logits), and we prove it by reporting ECE/Brier pre/post.

Why ONNX Runtime + XNNPACK on Pi, not PyTorch? ONNX Runtime with the XNNPACK provider is lighter and faster for ARM CPU inference; PyTorch's runtime footprint is heavy for an edge device. We verify ONNX↔PyTorch parity within 1e-4 so accuracy is unchanged.

Why the attention gate? A confident-but-wrong classifier is the dangerous failure. If Grad-CAM attention doesn't overlap the detected cells (IoU < 0.3), the model is "looking" at the wrong region, so we route to human review instead of trusting the score.

4. Threat model (federated setting)

- **Honest-but-curious server** — sees weight deltas, not images; optional Opacus DP (report ϵ , δ) bounds what a delta leaks.
- **Malicious client / poisoning** — mitigated by robust aggregation and per-client divergence monitoring (future: Krum/trimmed-mean).
- **Transport** — mutual TLS on the gRPC channel between clients and aggregator.
- **At rest** — AES-256-GCM for local artifacts; argon2id password hashing; Ed25519-signed reports; append-only hash-chained audit ledger.

5. Calibration & ECE — the crisp explanation

ECE bins predictions by confidence and measures the gap between average confidence and average accuracy per bin; a well-calibrated 80%-confidence bucket should be right ~80% of the time. Brier score is the mean squared error between predicted probabilities and one-hot truth. We report both before and after temperature scaling to prove the calibration actually improved.

6. DPDP alignment notes

Data minimisation (sample ID only, no PII by default), purpose limitation (screening triage), storage limitation (local, encrypted), and integrity (signed reports + audit chain). HemoSight is a research/portfolio project, not a certified device; real deployment needs ethics approval and clinician oversight.

7. Honest limitations to state proactively

"The numbers are on public datasets that don't match Indian PHC staining, so I report them as [BOOTSTRAP] and mark PENDING until trained. Real deployment needs [TARGET] validation. That honesty is a feature of the design, not a gap." Interviewers reward the candidate who names the boundary before they do.

HemoSight — Module Status

Author: M Sujith Sali, ISE Dept, VTU Karnataka. Legend: [WORKING] tests pass in CI · [PROTOTYPE] code complete, needs dataset/HW run · [PLANNED]

Module	Description	Status	Notes
0	Scaffold + determinism	[WORKING]	test_determinism passes; seed/egress/disclaimer utils done
1	Data contracts + preprocess + quality gate	[WORKING]	schema/quality/preprocess tests pass; download script + manifests + toy set
2	YOLOv8 detection (BCCD)	[PROTOTYPE]	trainer + ONNX export done; ONNX↔PyTorch parity test passes; mAP PENDING (make train-yolo)
3	WBC 5-class (Raabin)	[PROTOTYPE]	trainer + calibration + bootstrap CIs done; F1/ECE PENDING (make train-wbc)
4	Malaria binary (NIH)	[PROTOTYPE]	trainer + ROC/PR + calibration done; PENDING (make train-malaria)
5	Uncertainty + attention gate	[WORKING]	MC-Dropout + attention-gate tests pass; Grad-CAM wired
6	Federated (Flower)	[PROTOTYPE]	FedAvg/FedProx + non-IID sim done; strategy smoke test passes; curves PENDING (make federated-simulate)
7	Edge deploy (Pi 5)	[PROTOTYPE]	ONNX/XNNPACK infer + camera adapter + systemd done; latency PENDING (run on Pi)
8	FastAPI + React UI	[PROTOTYPE]	SSE stream + schemas + kn/hi/en UI + gauge + canvas done
9	Signed PDF + audit	[WORKING]	signed-PDF verify + audit-chain tests pass
10	Tests + docs bundle	[WORKING]	20 tests pass; README/MODEL_CARD/INTERVIEW_HANDBOOK/CONTRIBUTING done

Functional software status

The software runs **fully end-to-end today** on real (synthetic-data-trained) weights — no PENDING in the runtime path: `python -m scripts.train_toy -> pytest (22 pass) -> uvicorn api.main:app`. POST an image to `/analyze` and get populated cell counts, WBC differential, malaria flag, MC-Dropout uncertainty, attention-gate status, and a signed PDF.

Synthetic weights are for wiring/demo only; real accuracy needs Raabin-WBC / NIH Malaria (make train-*). Clinical metrics remain [BOOTSTRAP]/PENDING until those runs execute — by design, never fabricated.

Test summary: **22 passed** (determinism, schema, egress, quality, preprocess, attention, MC-dropout, audit, ONNX parity, signed PDF, federated smoke, **end-to-end pipeline, quality-gate rejection**).